

COURSE OUTLINE

ARCN 5005 - **Theory and Practice of Architectural Representation**

Fall 2024

Wednesday, 9:00 – 11:25 Room AA-209

1.0 - GENERAL INFORMATION**INSTRUCTORS**

Nicolas Arellano Risopatrón (he/him), BAS, PhD Candidate

TEACHING ASSISTANTS

N/A

OFFICE HOURS

Office hours will be online and flexible. We suggest that you make an appointment via email or talk to your instructor before or after class to make an appointment.

Wednesday 11:30 to 12:30 am (Online via Zoom link found on Brightspace or in person if required)

1. Weekly consultation with the instructor
2. All questions to be placed on GitHub issue. The process will be explained on the first class.
3. Questions shall be responded during office hours only. Students are encouraged to attend consultations.
4. consultation time to schedule the consultation. Instructor shall be present during office hours only if a consultation request has been received in the timeline requested and will only remain online as long as students have questions/discussions.
5. No course material questions will be answered via email, all questions relating to course material not brought up during a consultation must be submitted to the issues page, unless anonymity is required, in that case, email to the instructor will be permitted. (nicolasarellanorisop@cmail.carleton.ca).
6. All questions will be responded to in a manner that the entire class can benefit

CALENDAR DESCRIPTION

'Free-hand drawing' as a way of observing and understanding the world. Various media and techniques introduced through a wide range of studio exercises.

Includes: Experiential Learning Activities.

COURSE DESCRIPTION

The course reflects on the use of computers and computer coding for architectural representation, and it analyses the different ways in which coding can impact architecture.

It will explore the obstacles that architects must overcome in order to benefit from a fruitful relationship between computers and architecture.

Coding has the potential to further expand the current limits of our imagination. Computers are a tremendous contribution to our architectural exploration if we continue to explore innovative research that includes human input with computer logic and processing power to arrive at collaborative solutions.

Every class will be followed by a workshop that will teach students the skills to understand, modify, and to develop digital tools. This will give them awareness of the parts of a digital software in order to understand its biases and potential.

The full course description can be found in the course portal.

Topics:

- Digitalization
- Coding
- Open source software for architects
- 3d modelling
- Open BIM (IFC)
- Open GIS
- Collaboration techniques

PREREQUISITES & PRECLUSIONS

N/A

KEY STUDENT PERFORMANCE CRITERIA (SPC)

A1. Design Theories, Precedents, and Methods

A2. Design Skills

A3. Design Tools

COURSE AIMS AND LEARNING OUTCOMES

Upon completing this course students should be able to:

- 1) Use Open Source software for CAD, 3D modeling, BIM, photogrammetry, and GIS. Alternative to mainstream proprietary software
- 2) Introduction to basic coding and visual programming tools
- 3) Introduction to web development techniques
- 4) Web collaboration using GitHub.

2.0 - METHODOLOGIES, THEORIES AND PRINCIPLES

In person lectures, in person workshops, online lectures, visual presentations, supplementary readings and audiovisual resources, quizzes, and assignments.

3.0 ASSIGNMENTS, SCHEDULE, & ASSESSMENT

COURSE ASSIGNMENTS

1. Quizzes will be based on the material presented in the lectures.
2. Assignments will be based on the labs or workshops. They will be individual, and group based.

The assignments will be further articulated in the Project Brief

COURSE SCHEDULE

The full course schedule can be found in the course portal.

Week 01 – The art of computer graphics

Wednesday, September 4, 9:00 - 11:35am EST, Architecture Building - Room AA-209 (all classes are in-person only)

Week 02 – Programming and digitalization of the physical world

Wednesday, September 11, 9:00 - 11:35am EST, Architecture Building - Room AA-209

Week 03 – Representation tools and processes for the AEC

Wednesday, September 18, 9:00 - 11:35 am EST, Architecture Building - Room AA-209

Week 04 – Scripting

Wednesday, September 25, 9:00 - 11:35 am EST, Architecture Building - Room AA-209

Week 05 – Internet, Collaboration and the Open source movement

Wednesday, October 2, 9:00 - 11:35 am EST, Architecture Building - Room AA-209

Week 06 – Open source software for architects

Wednesday, October 9, 9:00 - 11:35 am EST, Architecture Building - Room AA-209

Week 07 – OS BIM Software: Bonsai (formerly BlenderBIM)

Wednesday, October 16, 9:00 - 11:35 am EST, Architecture Building - Room AA-209

Week 08 – Reading Week

Wednesday, October 23, No class

Week 9 – Web based 3D graphics (Online)

Wednesday, October 30, 9:00 - 11:35 am EST, Architecture Building – Online (Zoom link provided in Brightspace)

Week 10 – IFC to the browser

Wednesday, November 6, 9:00 - 11:35 am EST, Architecture Building - Room AA-209

Week 11 – Open Source GIS

Wednesday, November 13, 9:00 - 11:35 am EST, Architecture Building - Room AA-209

Week 12 – Web based OS mapping and GIS

Wednesday, November 20, 9:00 - 11:35 am EST, Architecture Building - Room AA-209

Week 13 – Advanced Toolbox - Going online, scalability and maintainability

Wednesday, November 27, 9:00 - 11:35 am EST, Architecture Building - Room AA-209

Week 14 – Final assignment workshop and assignment 3 presentations

Wednesday, December 4, 9:00 - 11:35 am EST, Architecture Building - Room AA-209

Week 15 – Final Assignment – Combine all together into a GIS/BIM platform

Date, time and place TBD - week of December 16th

CRITICAL DATES

The course calendar will be found here in the course portal.

Evaluation		Weight	
Quiz 1	15%	Week 5: October 2nd at 9:00	Online quiz (Brightspace) 20 minutes at the beginning of the class - It will cover contents of 4 first weeks
Assignment 1	10%	Week 7: October 16th at 9:00	Students' about page and assignment portal using HTML + CSS
Assignment 2	10%	Week 9: November 13th at 9:00	3d model using Blender, MeshRoom and Three.js
Quiz 2	10%	Week 12: November 27th at 9:00	Online quiz (Brightspace) 30 minutes at the beginning of the class - It will cover all course contents to date
Assignment 3	15%	Week 13: December 4th at 9:00	Custom map using Maplibre
Final Assignment	30%	Week 14: December 13th at 9:00	Combine all together into a GIS/BIM platform

ASSIGNMENT WEIGHT & GRADING

This course has multiple outputs spread throughout the duration of the term.

Quiz 1: content of first 4 weeks	15%
Quiz 2: It will cover all course contents to date	15%
Assignment 1: Students' about page and assignment portal using HTML + CSS	10%
Assignment 2: 3d model using Blender, MeshRoom and Three.js	10%
Assignment 3: Custom map using Maplibre	10%
Final Assignment: Combine all together into a GIS/BIM platform	30%
Attendance	10%

ATTENDANCE

Attendance to all lectures and labs is mandatory. It is your responsibility to be informed of decisions and announcements made during scheduled class time. Frequent unaccounted-for absences may result in a failing grade, whether or not assignments have been satisfactorily completed.

GRADING

For the grade in the "A" range, the instructor will have judged the student to have satisfied the stated objectives of the course in an outstanding to excellent manner; for the "B" range, in an above average manner; for the "C" range, in an average manner with C- being the lowest acceptable grade in the BAS - Design Core courses; for the "D" range, in the lowest acceptable manner in non-Core courses, and for "F", not to have satisfied the stated objectives of the course.

Grades will be assigned as A+ (90-100%), A (85-89%), A- (80-84%), B+ (77-79%), B (73-76%), B- (70-72%), C+ (67-69%), C (63-66%), C- (60-62%), D+ (57-59%), D (53-56%), D- (50-52%), F (0-49%) and ABS.

A grade of C- or better in each course of the BAS Core courses (ARCS) is required for a student to remain in *Good Standing*.

Please refer to the Undergraduate Calendar for regulations concerning grades and other program requirement information:

<https://calendar.carleton.ca/undergrad/regulations/academicregulationsoftheuniversity/grading/>

Regulations concerning grades and other program requirement information specific to the Architecture program can be found here:

<https://calendar.carleton.ca/undergrad/undergradprograms/architecturalstudies/>

**** Each grade will be based upon a comparison (1) with other students in the course and/or (2) with students who have previously taken the course and/or (3) with the instructor's expectations relative to the stated objectives of the course, based on his/her experience and expertise.**

FEEDBACK & EVALUATION PROCEDURES

For assignments: Every tutorial session is feedback. Students will receive additional feedback at the scheduled review dates and the work will be evaluated, and feedback given within two weeks of each assignment date.

For quizzes: Students will receive feedback at assessment points the same day.

EVALUATION CRITERIA

- **Self-directed work:** initiative in completing the required work and your engagement in each class,
- **Creativity, spatial imagination:** thoughtfulness, rigour and clarity of design thinking; intelligence and sophistication of architectural proposition, and how work compares to other students in the class and from previous years,
- **Process work, engagement and attitude:** preparedness at weekly tutorials/workshops and pin-ups; steady and improved elaboration of formal and spatial ideas (e.g. through sketches, drawings, and study models),
- **Application of technologies and skills:** knowledge, and ability of digital and analog techniques,
- **Completeness and craft of the work:** ability to meet deadlines.
- **Review preparedness:** the professionalism of presentation and quality of verbal communication and work presented.

Discretionary Evaluation Criteria – determined through class participation

- **Independence & self-directed work:** initiative and engagement in the studio work and the thoughtful evaluation and interpretation of all project requirements.
- **Review preparedness:** the professionalism of presentation and quality of verbal communication and work presented.
- **Attendance**

ACADEMIC INTEGRITY

It is a student's responsibility to know and understand Carleton University's Student Rights and Responsibilities and Academic Integrity policies:

<https://carleton.ca/secretariat/wp-content/uploads/Students-Rights-and-Responsibilities-Policy.pdf>

<https://carleton.ca/secretariat/wp-content/uploads/Academic-Integrity-Policy-2021.pdf>

- All academic work must be a student's own, completed through their intellectual and physical abilities.
- Understand the full breadth of what constitutes plagiarism; cite and reference all sources, including non-academic, video (tutorial) and film sources properly (Chicago style preferred).
- Unless explicitly stated or agreed to in writing, students may not use Artificial Intelligence tools (i.e., ChatGPT or Midjourney) to start, develop and/or complete an assignment. (see dedicated section about plagiarism)
- Collaboration or teamwork is encouraged. The assignment work and process must be available to be checked by classmates.
- Quizzes are meant to be kept private. Do not attempt to compare your quiz with your classmates since the questions/answers have been randomly ordered and differ from student to student.

INTELLECTUAL PROPERTY, COPYRIGHT AND FAIR DEALINGS

As a condition of participating in the course and for the purpose of academic evaluation, students will be required to upload in-progress and completed work to the instructor's desired online platform(s). It is expressly understood that any such records or copies of student work will be used for nonprofit presentation and for the purposes of this authorization, nonprofit presentation includes showing, screening, publication and releases or presentation as a public service by internet distribution, commercial broadcasting or publication in furtherance of course-specific and institutional learning objectives.

For reasons of intellectual property and copyright, please under no circumstances download course documents or presentations for distribution without first acquiring written permission of the author/instructor.

PLAGIARISM (Example #1: AI Tools Not Allowed)

Plagiarism is presenting, whether intentionally or not, the ideas, expression of ideas, or work of others as one's own, including content generated by AI tools. Plagiarism includes reproducing or paraphrasing portions of someone else's published or unpublished material, regardless of the source, and presenting these as one's own without proper citation or reference to the original source. Examples of sources from which the ideas, expressions of ideas or works of others may be drawn from include but are not limited to: books, articles, papers, literary compositions and phrases, performance compositions, chemical compounds, images, design, visual and textual precedents artworks, laboratory reports, research results, calculations and the results of calculations, diagrams, constructions, computer reports, computer code/software, material on the internet, and/or conversations.

Examples of plagiarism include, but are not limited to:

- any submission prepared in whole or in part, by someone else, including the unauthorized use of generative AI tools (e.g., ChatGPT or Midjourney). Unless explicitly stated or agreed to in writing, students may not use Artificial Intelligence tools to start, develop and/or complete an assignment;
- using ideas or direct, verbatim quotations, paraphrased material, algorithms, formulae, scientific or mathematical concepts, or ideas without appropriate acknowledgment in any academic assignment;
- using another's data or research findings without appropriate acknowledgement;

- submitting a computer program developed in whole or in part by someone else, with or without modifications, as one's own; and
- failing to acknowledge sources through the use of proper citations when using another's work and/or failing to use quotations marks.

Plagiarism is a serious offence that cannot be resolved directly by the course's instructor. The Associate Dean of the Faculty conducts a rigorous investigation, including an interview with the student, when an instructor suspects a piece of work has been plagiarized. Penalties are not trivial. They can include a final grade of "F" for the course.

CO-OPERATION OR COLLABORATION

While cooperation and collaboration among students is encouraged by the instructor to complete the assignments, students shall not cooperate or collaborate on academic work when the instructor has indicated that the work is to be completed on an individual basis. Failure to follow the instructor's directions in this regard is a violation of the standards of academic integrity. Unless otherwise indicated, students shall not cooperate or collaborate in the completion of a test or examination. Students are responsible for being aware of and demonstrating behaviour that is honest and ethical in their academic work (see www.carleton.ca/registrar). Instructors at both the graduate and undergraduate level are responsible for providing clear guidelines concerning their specific expectations of academic integrity (e.g. rules of collaboration or citation) on all course outlines, assignment and examination material.

Example #2: AI Tools Allowed (*Adapted from Mollick & Mollick, 2023; the authors gave their permission to use their language or adjust it to fit in one's own course*)

You are expected to use AI (e.g., ChatGPT and image generation tools) in this class. In fact, some assignments will require it. Learning to use AI is an emerging skill and the course will provide instructions on how to use them. The instructor can meet and help you with these tools during office hours or after class. Be aware of the limits of ChatGPT, such as the following:

- If you provide minimum-effort prompts, you will get low-quality results. You will need to refine your prompts in order to get good outcomes. This will take work.
- Do not trust anything ChatGPT says. If it gives you a number or fact, assume it is wrong unless you either know the answer or can check with another source. You will be responsible for any errors or omissions provided by the tool. It works best for topics you understand.
- AI is a tool, but one that you need to acknowledge using. **Please include a paragraph at the end of any assignment that uses AI explaining what you used the AI for and what prompts you used to get the results. Failure to do so violates the academic integrity policy.**
- Be thoughtful about when this tool is useful. Do not use it if it is not appropriate for the case or circumstance.

STATEMENT ON MENTAL HEALTH

As a University student you may experience a range of mental health challenges that significantly impact your academic success and overall well-being. If you need help, please speak to someone. There are numerous resources available both on- and off-campus to support you (see <https://carleton.ca/wellness/>):

EMERGENCY RESOURCES (ON AND OFF CAMPUS):

<https://carleton.ca/health/emergencies-and-crisis/emergency-numbers/>

CARLETON RESOURCES:

- Mental Health and Wellbeing: <https://carleton.ca/wellness/>
- Health & Counselling Services: <https://carleton.ca/health/>
- Paul Menton Centre: <https://carleton.ca/pmc/>
- Academic Advising Centre (AAC): <https://carleton.ca/academicadvising/>

- Centre for Student Academic Support (CSAS): <https://carleton.ca/csas/>
- Equity & Inclusivity Communities: <https://carleton.ca/equity/>

OFF CAMPUS RESOURCES:

- Distress Centre of Ottawa and Region: (613) 238-3311 or TEXT: 343-306-5550, <https://www.dcottawa.on.ca/>
- Mental Health Crisis Service: (613) 722-6914, 1-866-996-0991, <http://www.crisisline.ca/>
- Empower Me: 1-844-741-6389, <https://students.carleton.ca/services/empower-me-counselling-services/>
- Good2Talk: 1-866-925-5454, <https://good2talk.ca/>
- The Walk-In Counselling Clinic: <https://walkincounselling.com>

LATENESS

1% of the project grade will be deducted for every day of lateness. The first 1% is initiated immediately after the deadline.

OUT-OF-CLASS COMMITMENTS & WORKLOAD

Students are expected to work for +/- 3-4 hours for each weekly assignment outside of class time.

ANNOUNCEMENTS & CORRESPONDENCE

Please listen to all announcements and/or read all Brightspace correspondence carefully.

A growing frustration is the continual repetition of instructions, and request for additional clarification. An important ability in any profession is the thoughtful evaluation and interpretation of the project brief – this is a valuable skill all students need to learn and practice.

4.0 RESOURCES - COURSE COSTS

Computer and Software Requirements

Hardware:

1. It is expected that all students have access to a computer to complete course assignments. Unless discussed with the instructor, every student must bring their own laptop to every class.
2. It is expected that all students have access to a smartphone or digital camera.

Software: Design software to complete course assignments.

Free and /or Open source software will be sufficient to complete all the course assignments.

1. The students will be required to use the following programs:
Visual Studio Code: <https://code.visualstudio.com/download>
And its extensions
 - a. Live Server: <https://marketplace.visualstudio.com/items?itemName=ritwickdey.LiveServer>
 - b. Prettier: <https://marketplace.visualstudio.com/items?itemName=esbenp.prettier-vscode>
 - c. Material Icon Theme: <https://marketplace.visualstudio.com/items?itemName=PKief.material-icon-theme>
 - d. Console Ninja: <https://marketplace.visualstudio.com/items?itemName=WallabyJs.console-ninja>
 - e. Auto Import: <https://marketplace.visualstudio.com/items?itemName=steoates.autoimport>
2. Node js: <https://nodejs.org/en/download>
3. Blender: <https://www.blender.org/download/>
4. BonsaiBIM (formerly BlenderBIM): <https://blenderbim.org/download.html>
5. Meshroom: <https://alicevision.org/#meshroom>

Materials

N/A

IN-CLASS RESOURCES

1. The instructor will provide a list of recommended reading resources. The bibliography or the course can

- be found in the course portal
2. The instructor will provide the links of YouTube videos that the students must watch.

SELF-DIRECTED RESOURCES

Increasingly, student's desire additional computer resources and tutorials. Please review the school calendar for workshop dates

OUT-OF-CLASS COMMITMENTS & WORKLOAD

It will be recommended that the students participate in this webinar scheduled outside of class hours: Wednesday, September 20th – 13:00. That open company former ifc.js release.

<https://thatopen.com/f001-p01-launch-optin>

ACCOMMODATIONS

It is a student's responsibility to self-identify as needing accommodations and initiate a constructive dialogue with your instructor(s). If you require accommodations, please let your instructor(s) know early so that a plan can be developed and agreed upon in the beginning weeks of term.

STUDENT PERFORMANCE CRITERIA

For the purposes of accreditation, graduating students must demonstrate understanding or ability in the areas listed below, according to an established sequence. This table lists 24 STC for the professional programs sequence.

- A1. Design Theories, Precedents, and Methods
- A2. Design Skills
- A3. Design Tools
- A4. Program Analysis
- A5. Site Context and Design
- A6. Urban Design
- A7. Detail Design
- A8. Design Documentation

SECURITY AND SAFETY

- Avoid Working Alone After Hours · In Case of Emergency, Dial Extension 4444 from any campus phone.

EQUITY, DIVERSITY, AND INCLUSION COMMITMENT

The Azrieli School of Architecture and Urbanism is committed to foster and support equity, diversity, and inclusion. If you experience discrimination, micro-aggression, or other forms of racism, sexism, discrimination against 2SLGBTQ+, or disability, there are several pathways available for addressing this:

A) If you feel comfortable bringing this up directly with the faculty, staff or student who has said or done something offensive, we invite you, or a friend, to speak directly with this person. People make mistakes and dealing them directly in the present may be the most effective means of addressing the issue.

B) You can reach out to either the [Undergraduate office](#), [Graduate office](#), or Director ([Anne Bordeleau](#)). If you contact any of these people in confidence, they are bound to preserve your anonymity and follow up on your report.

C) You can choose to report centrally to the Department of Equity and Inclusive Communities (EIC). EIC can be reached by emailing equity@carleton.ca. More information on the functions and services of the EIC can be found here: <https://carleton.ca/equity/focus/>.

ACADEMIC ACCOMMODATION

You may need special arrangements to meet your academic obligations during the term. For an accommodation request the processes are as follows:

Academic consideration for medical or other extenuating circumstances:

Students must contact the instructor(s) as soon as possible, and normally no later than 24 hours after the submission deadline for course deliverables. *[Provide any additional information on your requirements for short-term informal accommodations. If you require supporting documentation for short-term considerations, you may only request the [Academic Consideration for Coursework form](#). You may not request medical notes or documentation.]*

Students should also consult the [Course Outline Information on Academic Accommodations](#) for more information. Detailed information about the procedure for requesting academic consideration can be found [here](#).

Pregnancy and Family-Status Related Accommodation: Please write to me with any requests for academic accommodation during the first few weeks of class, or as soon as possible after the need for accommodation is known to exist. For more details about the accommodation policy, visit the Equity and Inclusive Communities (EIC) website.

Religious obligation: write to me with any requests for academic accommodation during the first two weeks of class, or as soon as possible after the need for accommodation is known to exist. For more details [click here](#).

Academic Accommodations for Students with Disabilities: The Paul Menton Centre for Students with Disabilities (PMC) provides services to students with Learning Disabilities (LD), psychiatric/mental health disabilities, Attention Deficit Hyperactivity Disorder (ADHD), Autism Spectrum Disorders (ASD), chronic medical conditions, and impairments in mobility, hearing, and vision. If you have a disability requiring academic accommodations in this course, please contact PMC at 613-520-6608 or pmc@carleton.ca for a formal evaluation. If you are already registered with the PMC, please request your accommodations for this course through the [Ventus Student Portal](#) at the beginning of the term, and no later than two weeks before the first in-class scheduled test or exam requiring accommodation (*if applicable*). Requests made within two weeks will be reviewed on a case-by-case basis. For final exams, the deadlines to request accommodations are published in the [University Academic Calendars](#). After requesting accommodation from PMC, meet with me to ensure accommodation arrangements are made. Please consult the PMC website for the deadline to request accommodations for the formally-scheduled exam (if applicable).

Survivors of Sexual Violence

As a community, Carleton University is committed to maintaining a positive learning, working and living environment where sexual violence will not be tolerated, and where survivors are supported through academic accommodations as per Carleton's Sexual Violence Policy. For more information about the services available at the university and to obtain information about sexual violence and/or support, visit: <https://carleton.ca/equity/sexual-assault-support-services>

Accommodation for Student Activities

Carleton University recognizes the substantial benefits, both to the individual student and for the university, that result from a student participating in activities beyond the classroom experience. Reasonable accommodation will be provided to students who compete or perform at the national or international level. Write to me with any requests for academic accommodation during the first two weeks of class, or as soon as possible after the need for accommodation is known to exist. <https://carleton.ca/senate/wp-content/uploads/Accommodation-for-Student-Activities-1.pdf>

EXAM INFORMATION

e-Proctoring

[Scheduling and Examination Services \(SES\)](#) is offering e-proctoring for online and in-person digital exams this Fall. If you wish to use this service, apply via the [e-Proctoring application form](#) by September 20, and [mention or include a note about e-Proctoring in your course outlines](#). For more details, visit the SES [e-Proctoring overview page](#).

SES ADMINISTRATION OF IN-PERSON TESTS/EXAMS OUTSIDE OF CLASS TIME

Scheduling and Examination Services (SES) offers in-person exam administration for online courses without a fixed class time, or multi-section courses that require synchronized testing. Please specify in your course outline that assessments will occur on campus and outside of regular class time (which can include Friday evening, Saturday or Sunday). Services provided include conflict-free scheduling, space reservation, online exam creation if applicable, accommodation for PMC students, and proctoring. To use this service, [submit a request](#) by September 19.

Scheduling and Examination Services will reach out to you with finalized exam scheduling information by September 29. We strongly encourage those who are opting for assessments outside of class time to have them scheduled by SES in order to reduce conflicts and overloads for students.

For students studying remotely, varsity athletes, or those with extenuating circumstances, the flexibility of distance exams is available. Students interested should [apply by](#) September 22. More details can be found on the [Distance Exams overview page](#). If you have questions, contact OnlineExams@carleton.ca.

IMPORTANT DATES FOR 2024-25

A [full list of important dates](#) is available on the Calendar website. Please note that the academic withdrawal dates have changed recently. Consult the [Calendar website](#) for the most up to date information.

The following exceptions may apply:

- If it is necessary to hold a test after Friday, October 20 due to exceptional circumstances, it can only be scheduled on Saturday, October 21. This timing must be announced on the course outline that is distributed at the beginning of the term.
- Final exams for Early Fall undergraduate courses may be held on October 28 and 29.